NED University of Engineering and Technology
Department of Computer Science & Information Technology
FSCS - Midterm Examination Spring 2026
Logic Design and Switching Theory (CS-251)

Paper A | **Time: 90 minutes**

Roll No: [redacted] | **Marks: 20**

Q1: Answer the following questions: [CLO-1, Marks 10]

a. **Perform** subtraction using r's complement :

   i. $98.7654)_{10} - 987.654)_{10}$

b. **Demonstrate** by means of truth table the validity of Demorgan's Theorem for 3 input variables.

c. **Find** the Simplified expression using K-Map for the following function:
   F(A,B,C,D)=£(1,3,7,11,15)+ d(0,2,5)

d. **State** the difference between the following:

   i. Ripple Carry Adder and Carry Look ahead Adder.

   ii. Multiplexer and Demultiplexer.

Q2.**Design** a Full Subtractor Circuit . [CLO-2,Marks 3]

Q3.**Design** a 3 to 8 Active LOW enable input line decoder with Active LOW output.Also Show that this circuit is served as 1 x 8 Demultiplexer. [CLO-2,Marks 3]

Q4.**Apply** Boolean Algebra to simplify the following expressions: [CLO-2,Marks 4]

i. $X = [A'(B'C)' + A']' + [B(B'C)]'$

ii. $N = [(X_1' + X_1X_2) + (X_2' + X_1X_2')]'$